

Happje: Stimulating Collaborative Cooking for People with Dementia

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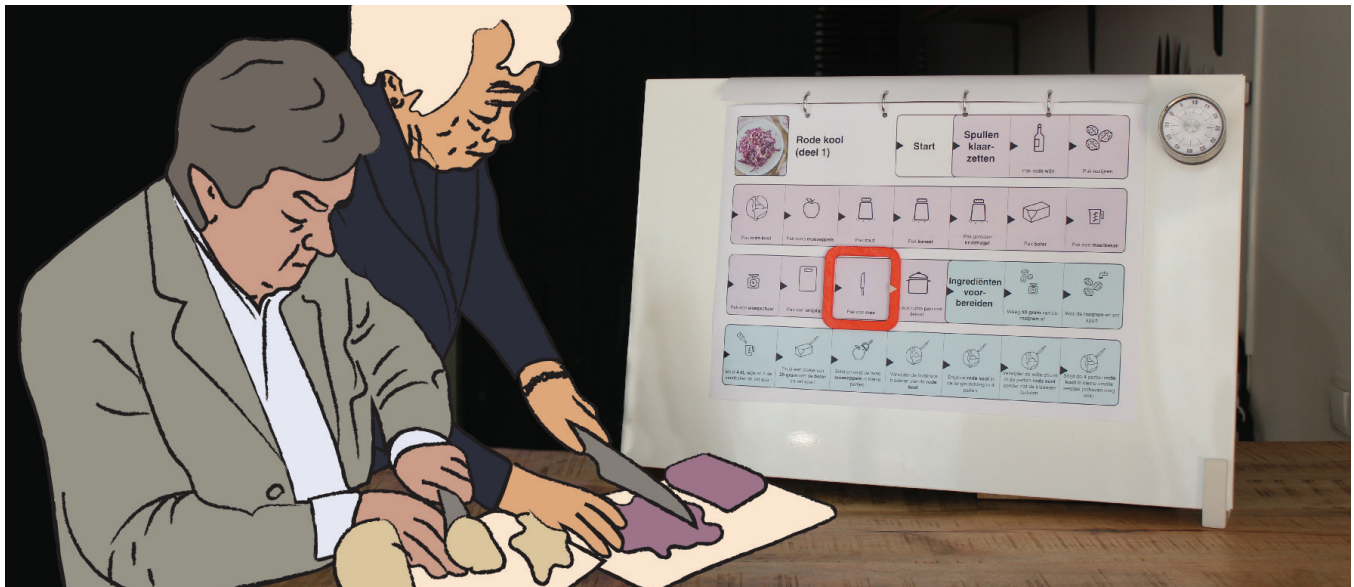


Figure 1: Final design Happje: a novel step-by-step cooking approach

ABSTRACT

People with dementia often experience difficulty performing daily activities, including cooking. Previous studies have focused on creating technological interventions for guiding people with dementia individually through cooking tasks and reducing errors. However,

collaborative cooking can often promote social engagement and increase self-esteem. This paper presents the iterative development of an easy-to-use recipe tool Happje with step-by-step instructions designed for and with people with dementia. Throughout the project, 13 iterations of the prototype were built and evaluated in the homes and meeting centres of people with dementia. Happje is designed to meet the cognitive abilities of people with dementia and promote collaboration. Moreover, Happje supports informal caregivers in creating personalised recipes. The present study emphasises the significance of utilising intuitive technology and fostering collaboration in daily living activities to promote autonomy and social engagement for individuals with dementia.

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CCS CONCEPTS

• **Human-centered computing** → **User centered design**; *Accessibility design and evaluation methods*.

KEYWORDS

dementia, cooking, collaboration, step-by-step

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1 INTRODUCTION AND BACKGROUND

The number of individuals with dementia continues to rise, from 55 million worldwide today to 139 million in 2050 [21]. And as a cure for dementia is still unavailable [21], the need for innovative technologies that support people with dementia in their daily challenges and enhance their quality of life becomes increasingly essential [3]. The decline in cognitive abilities associated with dementia can make activities of daily living, such as cooking, household chores, or personal hygiene, increasingly difficult. This increases the care load of the informal caregivers [9]. Moreover, it can result in a reduction in autonomy and self-esteem of the person with dementia [9]. It can lead to depression [12], which can, in turn, exacerbate functional disabilities [16].

Cooking is a particularly important daily activity for people with dementia, as it has been shown to promote social engagement, improve mood, and increase self-esteem [13, 14]. However, the cognitive demands of planning and executing cooking steps can make cooking challenging for people with dementia [11, 24]. Additionally, declines in reading abilities can make following written recipes more difficult [8], and reductions in short-term memory [2] make it challenging to remember cooking instructions. As a result, people with dementia often either lose interest in cooking because they find it too tiring, or their partners take over the cooking because they believe it is no longer up to standard [6].

1.1 Assistive Technologies for Cooking

Cooking activities have been receiving increasingly more attention in HCI research, including technology for menu planning, recipe recommendations, and cooking instructions, or to support people who are physically challenged in eating and cooking [30]. For example, the Ambient Kitchen, which uses sensors, cameras, and projectors to provide situated cooking prompts for people with dementia [20], and Augmented Reality applications that can give in-situ visual cues as a more comfortable and structured alternative to traditional printed recipes [28]. These studies have demonstrated the potential of technology to increase the independence (e.g., [1, 28]) and reduce errors (e.g., [29]) of people with dementia during cooking activities. However, there is limited research on how technology can support social, collaborative cooking, particularly for people with dementia. Furthermore, most studies were executed in laboratory environments and did not deal with the messy context of real lived-in homes [20].

1.2 Designing for Dementia in HCI

Over the years, research in the field of HCI has expanded from developing assistive technologies that solely address the cognitive impairment of people with dementia to designing for their psychosocial needs [7]. Studies have demonstrated the potential of technology and design to promote social engagement and agency in people with dementia [15] and to shift the dynamics of caregiving relationships to a more equal partnership between caregivers and care receivers [18].

The use of tangible prototypes or concrete design proposals in the early stages of the design process allows people with dementia to actively participate and critically reflect on the proposed solution [22]. This inclusive design approach not only provides insights into the everyday needs and context of people with dementia [17] but also contributes to their sense of usefulness and self-esteem [19].

In this paper, we present the design process of a recipe guide Happje ('bite' in Dutch) (Figure 1), to make cooking feel less intimidating for people with dementia while stimulating collaboration and social engagement.

2 METHOD

Before this study, a pilot phase involving interviews with three people with dementia and their care partners was conducted to gain insights into their cooking challenges. A participatory design approach was utilised, comprising 13 iterative cycles, until the time for the project had run out. The prototypes of these iterations (Table 1) were evaluated by 33 people with dementia, 13 formal caregivers, and 13 informal caregivers to ensure usability, usefulness, and acceptability [25, 27]. Iteration 11 was evaluated at a larger scale, described by Ruitenbunrg et al. [23].

2.1 Design process

Each iteration included the design of a prototype and an evaluation with the target group. The first prototype was based on a recipe with photos used in a cooking TV programme for people with dementia [26], and after thirteen iterations, we finished with the design Happje. Each iteration's prototype was evaluated during a cooking session (Figure 2) in the participants' homes (16 times), where they cooked with a partner or family member, or in local meeting centres (7 times), where multiple people with dementia gathered. Visiting real lived-in contexts and alternating between the two locations helped develop the design for a family home and a larger gathering and address different social relations like family, friends, and caregivers [20]. Throughout the project, 11 different recipes were used (e.g., cookies, cappuccino, beef stew) and chosen in consultation with the participants to fit their preferences. The recipes varied between main courses, drinks, and pastries to ensure Happje fit different types of cooking. During the evaluation, the first author observed and took photographs and field notes of the people with dementia and caregivers' interaction and cooking with the prototype. While eating the cooked dish, the first author invited the participants to evaluate the prototypes collaboratively. Based on these insights, a new prototype was built in the next iteration.

Participants were recruited through dementia support groups and meetings at local community centres. No official diagnosis was asked of the participants with dementia to not unnecessarily upset

Table 1: The thirteen prototypes with the location and participants of the evaluation

Label	Iterations	Evaluated at		Evaluated with		
		# Meeting centers	# Homes	# People with dementia	# Formal caregivers	# Informal caregivers
I1	Recipe booklet with pictures	1		3	2	
I2	Video recipe	1		3	2	
I3	Video recipe	1		4	2	
I4	Projections		1			
I5	Recipe with AI photos		1	1		1
I6	Boardgame recipe		1			
I7	Flowchart recipe		1	1		1
I8	Boardgame recipe with photos		1	1		1
I9	Zigzag boardgame recipe		1	1		1
I10	Training for writing recipes		1			
I11	Recipe on magnetic board	4	5	19	7	5
I12	Paper training writing recipes		4			4
I13	Final design					

**Figure 2: Evaluation at participant home (top) and meeting centre (bottom)**

them. Instead, we believed the caregiver when they stated a person had dementia. No distinction was made between the different types

or phases of dementia. Both male and female individuals with dementia participated in the study, aged 50 to 95. All participants with dementia lived at home alone or with an informal caregiver, and their cooking experience varied, ranging from no prior experience to having cooked daily. Some participants evaluated multiple iterations throughout the project, while others only evaluated the final one. The study was approved by the Human Research Ethics Committee of the Delft University of Technology, and before the study, written informed consent was obtained from caregivers. Verbal or written consent, depending on the participant with dementia’s ability [10], was obtained. Non-verbal cues were also observed during the participant’s participation for any indication that they wanted to withdraw from the study [4].

3 RESULTS

The insights from the design process are presented in five sections; step formulation, recipe appearance, step navigation, creating new recipes, and collaborative cooking. Iterations are referred to by their iteration number of Table 1.

3.1 Step formulation

As part of the cooking tool Happje, it was essential to ensure that the steps of a recipe were easy to understand and follow by people with dementia. In one of the first iterations, I2 (Figure 3), we presented multiple steps in one video with a voice-over to the participants. However, this approach confused the participants as they often had already forgotten the first task after watching the video. Additionally, the voice-over was often difficult to hear in the noisy kitchen environment. In I9, we presented the steps in written form, with each step mentioning one single action, which helped participants execute the step before forgetting. The steps were separated and grouped by colour-coded categories such as "Chopping" and "Mixing," which made the long list of steps feel less overwhelming. *"It is indeed very simple and very much step, step, step, which prevents you from feeling nervous. That is the thing with cooking; it makes you super nervous. And we are not getting nervous from this."*— participant with dementia I11. Moreover, each

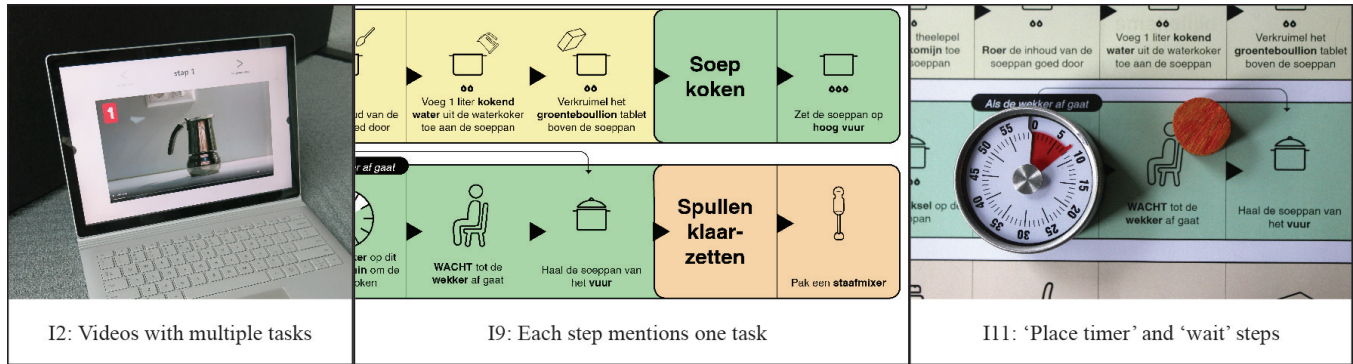


Figure 3: Step formulation iterations

step mentioned the number and allocation of the ingredients so participants did not have to recall earlier information. In iteration I11, we also included additional tasks such as a placeholder for a kitchen timer with instructions on what to do when it rings, and a "Turn off the stove" step, which further supported the person with dementia's safety and success in completing the recipe steps.

3.2 Recipe appearance

3.2.1 Visuals. Visual information in the recipe turned out to be crucial for the recipes, and multiple types of visuals were explored. Without any visuals (I7, Figure 4), the long list of steps felt overwhelming to participants as they felt they had to read every step before understanding the recipe. *"I prefer to have as little text as possible; the images have to be clear on their own."* – participant with dementia I7. Photos (I2) and videos (I3) were effective in explaining complicated cooking tasks but gave unrealistic expectations as participants expected to have the same tools and ingredients as depicted and would feel disappointed if their result did not match the one shown in the recipe. Later, we implemented icons (I9), which because of their more simplistic abstract look, were found to be more calming on the eyes and not set unrealistic expectations. The icons depicted a combined image of the tools, ingredients, and actions needed and helped participants understand every step of the process: *"Yes [we looked at the icons]. On there, you could exactly see what you had to do."* – participant with dementia I11.

3.2.2 Materials. Using videos shown on a laptop as used in I3 (Figure 5) was not preferred by people with dementia. Participants expressed discomfort with the use of unfamiliar and fragile technology in the kitchen *"I prefer not to have a step-by-step plan on the screen of a phone or laptop. I want nothing in the kitchen next to the stove that can break or be damaged."* – participant with dementia I8. Moreover, participants reported difficulties with the functionality of devices such as iPads *"The iPad screen always goes black after a while, and my husband does not understand how to fix it."* – partner of participant with dementia I9. Projections on the kitchen counter (I4) were quickly found impractical due to the need for a dark environment. In contrast, paper instructions (I1, I5–I13) were less intimidating for participants. They often resulted in the participants with dementia starting to cook independently without needing further stimulation from caregivers or researchers. *"Yes, I*

did [feel excited to start cooking when I saw the recipe]! I thought, gosh, fun! Especially because I am a bad cook, I thought, yes!" – participant with dementia I11. The paper recipes were placed on a magnetic board (I11) to take up less space, prevent the recipe from getting dirty, and let everyone who cooked see the steps.

3.3 Step navigation

3.3.1 Layout. The recipe layout was essential in facilitating task comprehension and navigation for people with dementia. In I1 (Figure 6), we found people with short-term memory issues could not remember which tasks they had completed and kept returning to the first step on the page. In I3, we presented one step at a time to prevent this issue, but this resulted in a lack of overview and hindered simultaneous task execution. Ultimately, in I11, we decided on a design that displayed all steps on a large sheet of paper. This improved the participants' comprehension of the overall process and allowed collaborating people to see different tasks simultaneously.

3.3.2 Marking Steps. To help participants track their progress in the recipe, markers for completed and current steps were introduced. Initially, an ink marker was proposed in I5 (Figure 7) to check off the completed steps manually, but this proved cumbersome to do after every task. A personal stamp (I8) helped more quickly mark the steps and indicated the current step by its position, but it was still found to be annoying to dab in ink every time. A magnetic pawn (I11) to only indicate the current position was experienced pleasantly by some participants. *"Normally, this is the moment that I look through all the steps and see what I have to do, but now I have marked where I left off, so I do not have to read through all of it again."* – participant with dementia I8. However, unlike with the stamp of I8, the pawn only allowed chronological step completion; participants could not randomly check off tasks which was challenging for cooks who enjoy improvising: *"I always just do whatever and now I have to stick to the protocol. And I have always been a free bird; that does not mean I cannot do it [follow the recipe chronologically], but it is more difficult for me."* – partner of a person with dementia I11. Participants with dementia also found it challenging to move the pawn consistently. *"I like it when I see you [moving and using the pawn]. But for myself, I think will, I will just continue. And I do not actually consciously think it; I just continue."* –

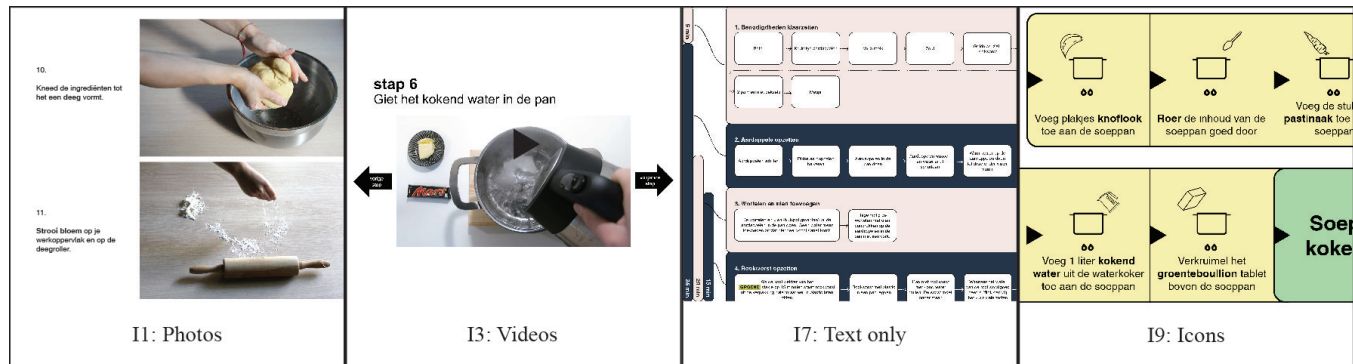


Figure 4: Recipe appearance iterations: visuals

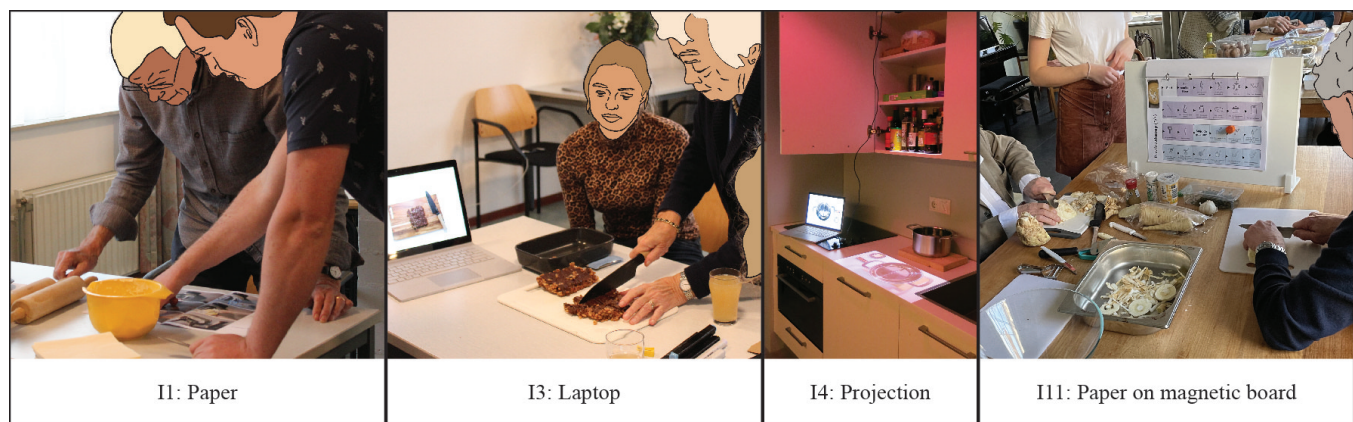


Figure 5: Recipe appearance iterations: materials

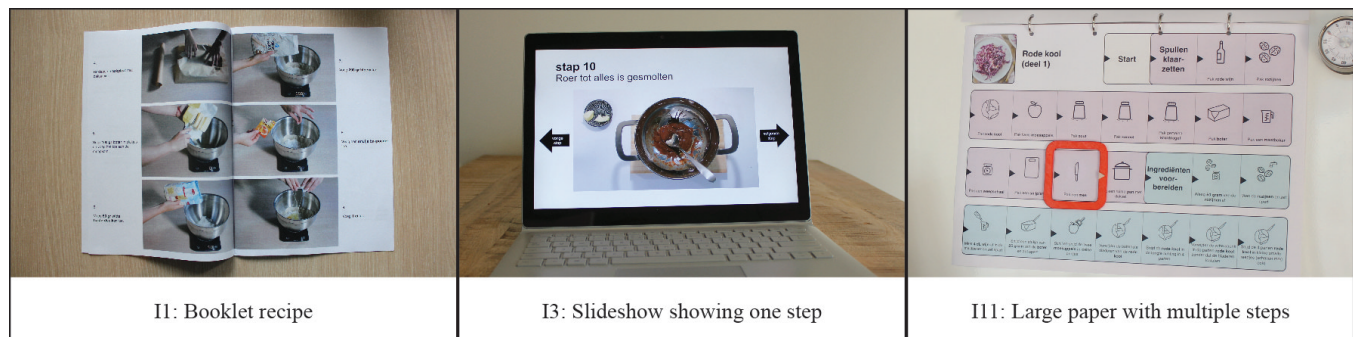


Figure 6: Step navigation iterations: layout

participant with dementia I11. And lastly, sometimes participants would take the pawn off the recipe and not remember where to put it back. *"At a certain moment, I saw that she took the pawn away and no longer knew where it belonged. She has to move it, but does she then still remember where it was before?"* - daughter of a person with dementia I11.

Despite these limitations, the indication of the current step did help others see which step the person with dementia was working on, thus allowing them to more easily check on the progress and

assist in the completion of all steps. For instance, when one friend looked at the recipe and saw the person with dementia had missed a step, she kindly reminded her of the fact. *"Sprinkle the chopped parsley over... Oh, we've already done that. And then [...]"* - a person with dementia. *"I think you are forgetting something."* - friend. *"Yes? Sprinkle the chopped parsley over the oven dish. Oh yes! [laughs]"* - person with dementia I11.



Figure 7: Step navigation iterations: marking steps

3.4 Creating new recipes

The participating caregivers stated they would rather use their own recipes than follow those of others. For example: *"We would rather have a good method to create my own recipes than a cookbook with predefined recipes. Everyone has different tastes and abilities change, so we would like to have the flexibility to do it ourselves."* - husband of a person with dementia I5.

3.4.1 Step definition. To address this need for personalised recipes, we developed a method for caregivers to translate their own recipes into our cooking tool. In I5 (Figure 8), we used a paper grid for the caregiver to write his steps, but this did not allow him to adjust or move them around. I12 used paper post-its for more flexibility but still required a lot of time to write by hand, and participants were not consistent in how they phrased the steps. Also, some participants still struggled with defining the steps because they found it difficult to imagine how the person with dementia would interpret them. *"Well, I did have to think carefully about that [defining the steps], whether I had added all the steps correctly. I am always a bit of a doubter, because I know what I want to bring across, but I do not know, because you are not face-to-face, how that will translate to another person."* - friend of a participant with dementia I12. Therefore, in I13, we proposed a digital interface for step definition, which would provide recommendations for missing information and a database of pre-existing adjustable recipes for caregivers with less time or motivation.

3.4.2 Adding visuals. Additionally, the recipes were supported by visuals. Initially, we asked a caregiver to take pictures of each step but found this was time-consuming and still provided relatively unclear imagery (I8, Figure 9). We then explored the use of AI-generated visuals based on the textual instructions. AI photos (I5) required an extensive database of accurate images and did not yet generate correct and recognisable images. Instead, we proposed using AI-generated icons (I9), which could be sourced from copyright-free sources on the web.

3.5 Collaborative cooking

Collaborative cooking was found to have many positive effects during the study. Cooking with other people with dementia created a fun social setting where participants were less concerned about making mistakes. For instance, one informal caregiver said: *"He is not a cook."* - partner of a person with dementia, to which the

person with dementia replied: *"That does not matter, we are simply doing it."* - person with dementia I3. However, when cooking with their partner, the partner often focused more on avoiding mistakes than having fun. *"Yes, I was indeed checking a little bit. Whether everything was going in at the right moment and that the fire was at some point still on without any oil in the pan. And also checking a little if she was doing it according to the steps."* - daughter of a participant with dementia I11. On the other hand, the collaborative cooking activity also made caregivers realise how capable people with dementia still were. *"It is good to be aware of that [that the participants could still cook] because it sometimes sneaks in to always take over things, while they are capable of doing a lot themselves."* - caregiver at meeting centre I11.

All groups in I11 reported enjoying collaborating while cooking and expressed a desire to do so again. *"I really enjoyed it! [...] I don't do it [cooking] at home anymore, but that does not mean I no longer enjoy it. I have baked so much in my life."* - person with dementia I11. One participant with dementia also highlighted that collaborative cooking was a great way to keep all her peers involved. *"Yes [we would like to cook this way again]. Because it is a good way to keep everyone engaged, so they can learn something new. Because you are not just doing it for you, but also for each other."* - participant with dementia I11.

4 DISCUSSION

Our study aimed to design a tool, Happje, to support people with dementia in collaborative cooking. Through thirteen iterations with prototypes and evaluations, we identified key design decisions that informed the final design and might inform future design work in the field of HCI concerning designing for people with cognitive impairment or collaborative cooking.

The design of Happje, a cooking tool for people with dementia, includes step-by-step paper instructions and a digital interface for caregivers to create personalised recipes. Our findings align with previous research on the efficacy of step-by-step instructions for people with dementia in daily tasks such as cooking [29]. We found that presenting one task per step and grouping tasks by type was beneficial for task execution and reducing cognitive overload. Using icons at each step effectively provided visual information of the actions and required components [28] while avoiding unrealistic expectations caused by realistic imagery [28]. The study aligns with prior research indicating that unfamiliar technology, such as screens or AR, in the kitchen can increase anxiety among

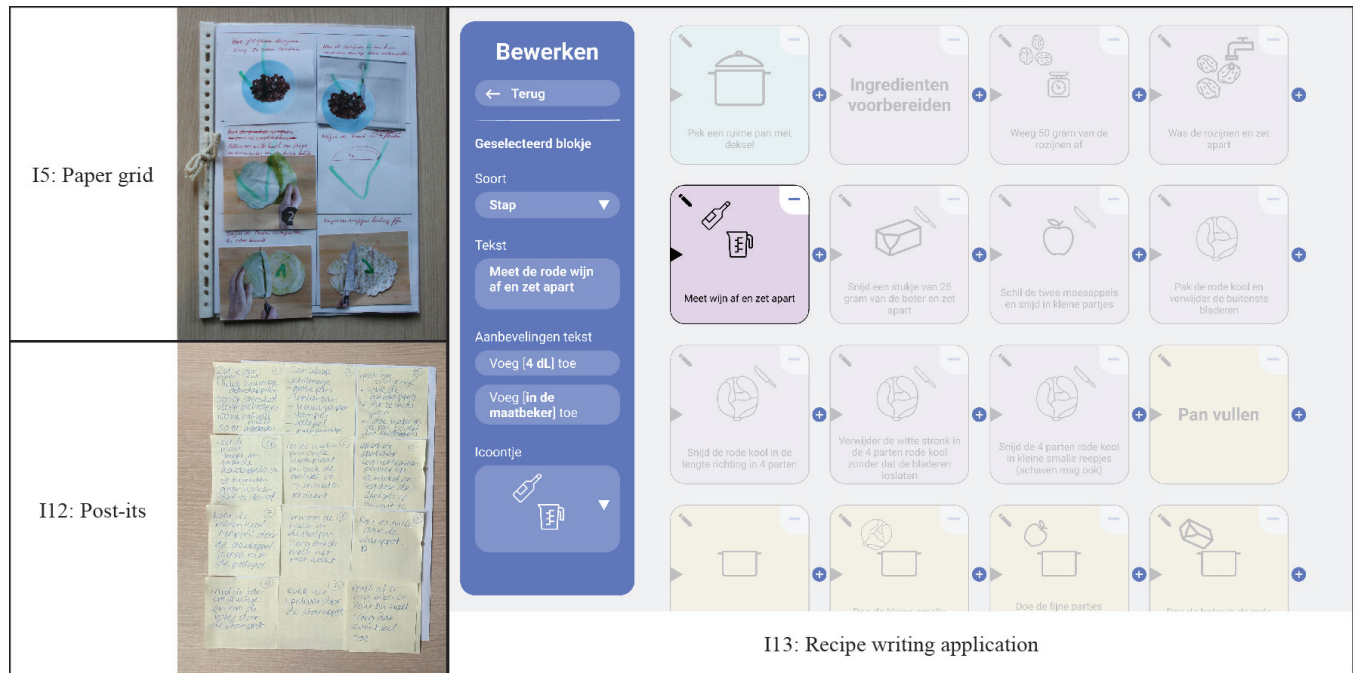


Figure 8: Creating new recipe iterations: step definition

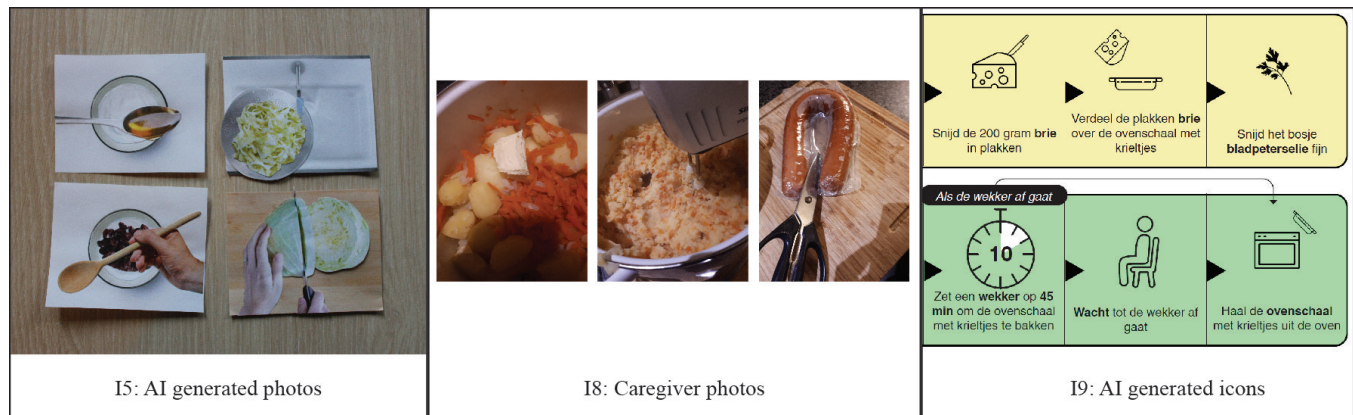


Figure 9: Creating new recipe iterations: adding visuals

individuals with dementia and decrease their willingness to acquire it in daily use [28]. Thus, we emphasise the importance of using familiar technology that is unlikely to be damaged in a kitchen to decrease intimidation for people with dementia. Breaking up a complicated task into small steps with easy-to-understand visuals and a well-organised layout was found to reduce feelings of overwhelm and discomfort, leading to increased autonomy and self-efficacy as participants with dementia would initiate the activities without stimulation needed from a caregiver. The study highlights the importance of easy-to-use and intuitive technology in reducing the intimidation factor for people with dementia, thereby mitigating their tendency towards apathy [5] in engaging in complex activities of daily living.

Our findings align with previous research [13, 14] that suggests collaborative cooking can be perceived as an enjoyable social activity for people with dementia. We observed that participants felt less anxious about making mistakes during the collaborative cooking experience. To facilitate collaboration and address the tendency for participants to deviate from the chronological order of steps, as previously reported in literature [28], clear markers are essential to indicate completed and ongoing steps, and all steps should be made visible to all participants. The study highlights the potential benefits of collaborative activities of daily living for people with dementia, as they can promote social engagement and reduce the fear of making mistakes. However, when engaging in collaborative

activities with informal caregivers, it is essential to shift the focus from minimising errors to fostering enjoyable experiences.

Consistent with previous research [1, 28], we identified recipe personalisation as crucial to ensure suitability for people with dementia by addressing their preferences and instructional needs and tapping into personal memories. Caregivers have deemed the most appropriate authors for these personalised recipes, although they can better be supported in how they write cooking instructions. The application design of Happje allows caregivers to define the cooking steps, automatically generates corresponding icons, and includes a database of easily adjustable recipes. Further research should evaluate the effectiveness of the Happje writing application for caregivers and its adaptability to the changing abilities of people with dementia over time. Cultural appropriateness outside of the Netherlands and the potential applicability of Happje in other daily activities (such as laundry) should also be investigated in future research.

5 CONCLUSION

In conclusion, the present study aimed to design a non-intimidating tool, Happje, that supports individuals with dementia in their daily cooking activities while collaborating with others with dementia or caregivers. The results of this study indicate that using intuitive step-by-step instructions and collaborative execution of tasks can effectively reduce the perceived intimidation of daily activities, thereby promoting autonomy and engagement among people with dementia. Additionally, the study found that caregivers are often motivated to write personalised recipes but require support in doing so. Future research should focus on developing and evaluating an application that assists caregivers in writing these recipes.

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